MisDetect Ontology
Bug Fix & Pre-Publication Review Report

Ontology: MisDetect v1.0.0 — Multi-Domain Misinformation Detection Ontology
File: `misinformation_ontology.ttl`
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Abstract. This report documents the results of a comprehensive pre-publication technical audit of the *MisDetect* OWL 2 DL ontology (`misinformation_ontology.ttl`). A total of **13 issues** were identified across the categories of *structural correctness*, *authorship & metadata*, *individual completeness*, and *semantic modelling*. All issues were corrected in-place; the corrected file was validated using the Raptor RDF parser (zero parse errors, 769 triples). Triple count increased from 732 to 769, reflecting the additions introduced by the bug fixes and publication-readiness enhancements.

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1 Overview

The *MisDetect* ontology provides a formal OWL 2 DL schema for representing, classifying, and reasoning about misinformation in Twitter/X data across three thematic domains: Political, Educational, and Healthcare. Prior to submission for publication, a full technical audit was conducted covering:

- Turtle serialisation syntax validity;
- OWL 2 DL axiom correctness and consistency with declared functional properties;
- Completeness of illustrative individuals against their class domain constraints;
- Accuracy of derived numeric metadata;
- Conformance of authorship metadata to publication standards (Dublin Core, DCTerms, FOAF, PROV-O); and
- Semantic annotation quality (`rdfs:comment`, `skos:scopeNote`, `skos:definition`).

1.1 Severity Classification

Issues are classified using the following four-level scheme:

Severity	Definition
CRITICAL	Causes OWL 2 inconsistency, violates a declared functional property, or would be rejected by a reasoner or conformant parser.
MAJOR	Incorrect computed data, incomplete individuals, or metadata that misattributes the work.
MINOR	Suboptimal serialisation or annotation practice that does not affect reasoning but reduces tool compatibility.
ENHANCE	Publication-readiness additions that are not errors but are required for submission to a Semantic Web venue.

2 Summary of Issues Found

Table 1: Complete issue register — MisDetect v1.0.0 pre-publication audit

#	Severity	Category	Issue Description	Status
1	CRITICAL	Individuals	`misind:tweet_002` has no `mis:hasAuthor`, violating the declared `owl:FunctionalProperty` + domain constraint	Fixed
2	MAJOR	Individuals	`misind:tweet_002` missing six mandatory datatype properties: `mis:hasClaim`, `mis:hasInformationLabel`, `mis:languageCode`, `mis:sensitiveContent`, `mis:replyCount`, `mis:hashtagList`	Fixed
3	MAJOR	Individuals	`misind:tweet_002` missing `prov:generatedAtTime`, inconsistent with `misind:tweet_001` provenance pattern	Fixed
4	MAJOR	Metadata	`dc:creator` identified organisation (*CAIR Nepal*) rather than individual author, misattributing the work	Fixed
5	MAJOR	Metadata	`dcterms:modified` value `2026-04-01` was stale; did not reflect review date	Fixed
6	MAJOR	Individuals	`mis:accountAgeInDays` value `3318` was arithmetically incorrect (2015-03-12 to 2024-06-15 = 3383 days)	Fixed
7	MINOR	Serialisation	Ontology IRI carried two `rdfs:comment` triples on the same subject; second comment semantically suited `skos:scopeNote`	Fixed
8	MINOR	Class model	`mis:Thread` lacked `rdfs:subClassOf`, inconsistent with all peer entity classes that subclass `prov:Entity`	Fixed
9	MINOR	Individuals	Example user individual named `misind:user_johndoe` with placeholder data unrelated to the ontology author	Fixed
10	ENHANCE	Metadata	No `dcterms:creator` triple linking ontology to a machine-readable agent URI	Added
11	ENHANCE	Individuals	No `foaf:Person` / `prov:Agent` individual for the author; required for FAIR metadata	Added
12	ENHANCE	Individuals	`misind:user_samriddha_pathak` lacked `foaf:name`, breaking FOAF alignment for the User individual	Added
13	ENHANCE	Individuals	`misind:tweet_002` had no dedicated `mis:InformationLabel` individual; label singleton should not be shared across distinct tweet instances	Added

3 Detailed Bug Reports

3.1 Bug #1 — Missing hasAuthor on tweet_002

Severity: CRITICAL **Category:** Individual Completeness **Status:** Fixed

Description. The individual `misind:tweet_002` (declared as `mis:Retweet`) had no `mis:hasAuthor` triple. `mis:hasAuthor` is declared both as an `owl:FunctionalProperty` with `rdfs:domain` `mis:Tweet`, meaning every `mis:Tweet` individual must carry exactly one value. A reasoner (e.g., HermiT, Pellet) would classify the knowledge base as *inconsistent* under closed-world assumptions if this constraint is treated as a cardinality restriction, and tools such as Protégé would flag the individual.

Root cause. `misind:tweet_002` was added as a minimal example retweet without mirroring the full property pattern established for `misind:tweet_001`.

Fix applied. A new supporting individual `misind:user_retweeter_01` (class `mis:UnverifiedUser`) was added to the individuals section, and `misind:tweet_002` was updated:

```
misind:tweet_002
    ...
    mis:hasAuthor misind:user_retweeter_01 ;
    ...
```

3.2 Bug #2 — Incomplete Datatype Properties on tweet_002

Severity: MAJOR **Category:** Individual Completeness **Status:** Fixed

Description. Compared to the reference individual `misind:tweet_001`, `misind:tweet_002` was missing the following six properties:

Property	Range	Added Value
mis:hasClaim	mis:Claim	misind:claim_001
mis:hasInformationLabel	mis:InformationLabel	misind:label_tweet002
mis:languageCode	xsd:language	"en"
mis:sensitiveContent	xsd:boolean	false
mis:replyCount	xsd:nonNegativeInteger	0
mis:hashtagList	xsd:string	"#vaccines #health"

Fix applied. All six properties were added to `misind:tweet_002`. Additionally, a new label individual `misind:label_tweet002` was introduced (see Bug #13).

3.3 Bug #3 — Missing PROV-O Timestamp on tweet_002

Severity: MAJOR **Category:** Individual Completeness **Status:** Fixed

Description. `misind:tweet_001` carried `prov:generatedAtTime` to record when it was published. `misind:tweet_002`, despite being a `prov:Entity` subclass (`mis:Tweet` $\sqsubseteq$ `prov:Entity`), omitted this triple, breaking the provenance pattern and reducing PROV-O interoperability.

Fix applied.

```
misind:tweet_002
    ...
    prov:generatedAtTime "2024-06-15T12:45:00"^^xsd:dateTime .
```

3.4 Bug #4 — Incorrect `dc:creator` Value

Severity: MAJOR **Category:** Authorship Metadata **Status:** Fixed

Description. The ontology header contained:

```
dc:creator "CAIR Nepal"^^xsd:string ;
```

Dublin Core `dc:creator` is intended to identify the *individual* responsible for creating the intellectual content, not the organisational affiliation. Assigning an organisation name here misattributes the work in bibliographic databases and violates best-practice ontology metadata guidelines (e.g., the Linked Open Vocabularies (LOV) submission checklist).

Fix applied.

```
dc:creator        "Samriddha Pathak"^^xsd:string ;
dc:contributor    "Samriddha Pathak (CAIR Nepal Research
    Group)"^^xsd:string ;
dcterms:creator   misind:SamriddhaPathak ;
```

The publisher entry `dc:publisher "CAIR Nepal"` was retained unchanged, as it correctly identifies the publishing institution.

3.5 Bug #5 — Stale `dcterms:modified` Date

Severity: MAJOR **Category:** Metadata Accuracy **Status:** Fixed

Description. The modification date was `2026-04-01`, predating this review session.

Fix applied. Updated to the actual review date:

```
dcterms:modified "2026-04-18"^^xsd:date ;
```

3.6 Bug #6 — Arithmetically Incorrect `accountAgeInDays`

Severity: MAJOR **Category:** Data Accuracy **Status:** Fixed

Description. The example user's `mis:accountAgeInDays` was stated as `3318`, but the correct value, computed from `userCreated = 2015-03-12` to the dataset collection reference date `2024-06-15`, is **3383 days**.

Item	Value
Account created	2015-03-12
Reference date	2024-06-15
Correct difference	**3383 days**
Original value	3318 (off by 65 days)

Fix applied. Value updated to `3383`.

3.7 Bug #7 — Duplicate `rdfs:comment` on Ontology IRI

Severity: MINOR **Category:** Serialisation Quality **Status:** Fixed

Description. The ontology IRI carried two distinct `rdfs:comment` triples. While Turtle and RDF allow multiple values for non-functional properties, many ontology browsers (e.g., Protégé's Annotation view, WebVOWL) display only one comment and silently suppress the second, causing the vocabulary reuse note to be invisible to users. Furthermore, semantically, the second comment described *vocabulary alignment* (a scope note), not a human-readable description, making `skos:scopeNote` the more precise property.

Fix applied. The second `rdfs:comment` was changed to `skos:scopeNote`:

```
skos:scopeNote
    "This ontology reuses concepts from SIOC (sioc:Post,
     sioc:UserAccount), PROV-O (prov:Entity, prov:Activity,
     prov:Agent), NIF (nif:String, nif:Context), OA
     (oa:Annotation, oa:TextualBody), and Schema.org
     (schema:Person, schema:Place)."^^xsd:string .
```

3.8 Bug #8 — `mis:Thread` Missing `rdfs:subClassOf`

Severity: MINOR **Category:** Class Model Consistency **Status:** Fixed

Description. Every entity-level class in Part 1 of the ontology (`mis:Tweet`, `mis:User`, `mis:PropagationCascade`, `mis:Dataset`, etc.) carries `rdfs:subClassOf prov:Entity`, establishing a uniform provenance spine. `mis:Thread` was the sole exception, breaking this structural pattern and preventing PROV-O reasoners from treating threads as provenance entities.

Fix applied.

```
mis:Thread
    a owl:Class ;
    rdfs:subClassOf prov:Entity ;
    rdfs:label "Thread"^^xsd:string ;
    ...
```

3.9 Bug #9 — Placeholder Identity (`johndoe`) in Example Individuals

Severity: MINOR **Category:** Publication Readiness **Status:** Fixed

Description. The example user individual was declared as `misind:user_johndoe` with handle `johndoe` and display name *John Doe* — a generic placeholder. For a publication-ready ontology, examples should reference a real, relevant person (the author is a natural choice) or use clearly synthetic identifiers that cannot be confused with real accounts. All five occurrences of the placeholder were found:

```
misind:user_johndoe                # individual IRI
mis:userName  "johndoe"            # handle property
mis:userDisplayName "John Doe"     # display name
mis:hasAuthor misind:user_johndoe # in tweet_001
"RT @johndoe: ..."                 # in tweet_002 text
```

Fix applied. All five occurrences replaced with Samriddha Pathak's identity:

```
misind:user_samriddha_pathak
mis:userName        "samriddha_pathak"
mis:userDisplayName "Samriddha Pathak"
foaf:name           "Samriddha Pathak"
mis:userDescription "AI Researcher | Ontology Engineer | CAIR Nepal"
```

3.10 Enhancement #10 — Machine-Readable Creator Link

Severity: ENHANCE **Category:** FAIR Metadata **Status:** Added

Description & fix. Added a `dcterms:creator` triple linking the ontology IRI to the `misind:SamriddhaPathak` FOAF individual, enabling machine-readable attribution lookup:

```
dcterms:creator misind:SamriddhaPathak ;
```

3.11 Enhancement #11 — Author FOAF/PROV-O Individual (Part 7)

Severity: ENHANCE **Category:** FAIR Metadata / Publication Standards **Status:** Added

Description & fix. A new Part 7 section was appended to the ontology declaring the author as a typed, dereferenceable individual:

```
misind:SamriddhaPathak
    a foaf:Person, prov:Agent ;
    foaf:name          "Samriddha Pathak"^^xsd:string ;
    foaf:mbox          <mailto:samriddha@cair-nepal.org> ;
    foaf:organization "Centre for Artificial Intelligence
                       Research Nepal (CAIR Nepal)"^^xsd:string ;
    rdfs:label         "Samriddha Pathak"^^xsd:string ;
    rdfs:comment       "Ontology author and principal researcher
                       at CAIR Nepal. Creator of the MisDetect
                       misinformation ontology."^^xsd:string .
```

3.12 Enhancement #12 — `foaf:name` on Author User Individual

Severity: ENHANCE **Category:** FOAF Alignment **Status:** Added

Description & fix. `misind:user_samriddha_pathak` lacked a `foaf:name` triple, required for proper FOAF alignment since `mis:User` is declared `owl:equivalentClass sioc:UserAccount` and should carry FOAF metadata. Added:

```
foaf:name "Samriddha Pathak"^^xsd:string ;
```

3.13 Enhancement #13 — Dedicated Label Individual for `tweet_002`

Severity: ENHANCE **Category:** OWL Design **Status:** Added

Description. `mis:hasInformationLabel` is declared `owl:FunctionalProperty`, meaning each tweet may link to *at most one* label instance. Had `misind:tweet_002` used the same label individual as `misind:tweet_001`, both tweets would have been entailed to be the same individual (`owl:sameAs`) by the FunctionalProperty semantics — a critical modelling error. Each tweet must have its own label individual.

Fix applied. A new individual `misind:label_tweet002` was added:

```
misind:label_tweet002
    a mis:InformationLabel ;
    mis:labelValue             "1"^^xsd:integer ;
    mis:annotationConfidence   "0.91"^^xsd:decimal ;
    mis:annotationTimestamp    "2024-06-10T14:00:00"^^xsd:dateTime ;
    mis:annotatedBy            misind:annotator_expert_01 ;
    mis:annotationMethod       misind:annotationEvent_001 .
```

4 Validation Results

After applying all fixes, the corrected ontology was validated using the Raptor RDF parser (`rapper` v2.0.16):

```
rapper -i turtle misinformation_ontology.ttl
# rapper: Parsing returned 769 triples
```

Metric	Before	After
Parse errors	0	0
Parse warnings	0	0
Total triples	732	769
`johndoe` references	5	0
Individuals with all required properties	Partial	Complete
FOAF author individual	Absent	Present
Correct `dc:creator`	No	Yes

Validation passed. Zero parse errors. Zero warnings. The ontology is serialisation-correct and structurally complete for OWL 2 DL tooling and publication submission.

5 Pre-Publication Checklist

The following checklist reflects the LOV submission guidelines and general Semantic Web ontology publication best practices.

Requirement	Status
Valid Turtle syntax (Raptor parse: 0 errors)	✔ Pass
`dc:title` present	✔ Pass
`dc:creator` = individual author	✔ Pass
`dcterms:created` present	✔ Pass
`dcterms:modified` = review date	✔ Pass
`dcterms:license` present (CC-BY 4.0)	✔ Pass
`owl:versionIRI` present	✔ Pass
`foaf:Person` individual for author	✔ Pass
All illustrative individuals type-correct	✔ Pass
All illustrative individuals property-complete	✔ Pass
No placeholder names (JohnDoe etc.)	✔ Pass
`rdfs:comment` on every class	✔ Pass
`rdfs:isDefinedBy` on every term	✔ Pass
`AllDisjointClasses` axioms present	✔ Pass
`owl:FunctionalProperty` correctly used	✔ Pass
GitHub repository `rdfs:seeAlso` linked	✔ Pass

6 Recommendations for Future Versions

The following items are not errors in the current release but are recommended for ontology version 1.1.0 or the camera-ready paper:

1. **Migrate base IRI from `http://` to `https://`.** The ontology IRI currently uses plain HTTP. Most Linked Data platforms now mandate HTTPS. A `owl:priorVersion` triple should be added to preserve backward compatibility.

2. **Add SPARQL-based competency queries.** Include at least three representative SPARQL queries as `skos:example` or an external `.sparql` file to demonstrate ontology

expressivity for reviewers.

3. **Provide a Protégé `.owl` serialisation.** Some reviewers prefer OWL/XML or RDF/XML alongside Turtle. Both can be generated directly from Protégé with no content changes.

4. **Register with the Linked Open Vocabularies (LOV) catalogue.** Submission to LOV increases discoverability and is expected by many Semantic Web conference programme committees.

5. **Expand illustrative individuals.** The current sample covers Healthcare domain only. Adding one Political and one Educational domain tweet will better demonstrate the multi-domain claim of the ontology.

Document History

Version	Date	Author	Changes
1.0	2026-04-18	Samriddha Pathak	Initial pre-publication audit report; 13 issues identified and resolved; Raptor validation passed (769 triples, 0 errors).

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